Person 3 - Phases 2-4 (T3.4 - T3.8): Subscriptions, Tax, Billing, Invoices

Works on top of Phase 1 (catalog + plans). Package: `com.subscriptiontracker` (change to yours).

Files

- `db/migration/V4__subscriptions_tax_billing.sql` (PostgreSQL; needs V3 from Phase 1)
- `subscription/` QuotationTemplate, Subscription (state machine), SubscriptionService, controllers, CycleMath
- `tax/` TaxRule, TaxService (integer tax maths), TaxController
- `billing/` Invoice, InvoiceLine, JobRun, SubscriptionBiller, BillingRunService, controllers, hooks

Flow

create draft -> /quotation?templateId= -> /confirm -> /activate -> POST /api/billing/runs -> invoice DRAFT
-> /api/invoices/{id}/confirm -> (payments module) PAID. Cancel only if no payments.

Endpoints (writes = ADMIN)

POST/GET /api/subscriptions, GET /{id}, POST /{id}/quotation|confirm|activate|pause|resume|renew|close
GET/POST/PUT /api/quotation-templates
GET/POST/PUT/DELETE /api/tax-rules
POST/GET /api/billing/runs, GET /api/billing/runs/{id}
GET /api/invoices, GET /{id}, GET /{id}/print, POST /{id}/confirm|cancel

Hooks for other people

- Person 4 (discounts): implement `billing.DiscountService` as a Spring bean. Until then discount = 0.

- Payments: implement `billing.PaymentLookup` (paidAmount) and call `InvoiceService.markPaid(id)`.
- customers: `customer_id` is a plain Long, no foreign key (their table name unknown).

Config (application.properties, all optional)

billing.cron=0 0 2 * * * # daily 02:00. Use "-" to disable the scheduler
billing.due-days=15 # invoice due date = period start + N days

Notes

- Money is long paise everywhere. Tax = (amount * bps + 5000) / 10000, once per line, after discount.
- Idempotent: UNIQUE(subscription_id, period_start); re-running a billing run creates nothing new.
- One run bills one period per subscription. Invoice numbers come from a DB sequence (rollbacks leave gaps).
- Sequences use PostgreSQL `nextval`. For MySQL replace nextNumber() with a counter table.